MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Mobile Shop Lifecycle Management System

Submitted in partial fulfillment for the award of

Diploma in Computer Engineering

By

Ansari Md Sohail Md Husain

Chuger Hanish Sunil

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DEPARTMENT OF COMPUTER ENGINEERING

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(Institute Code : 0059)

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Abstract

The Mobile Shop Management System is developed to modernize and streamline the complete lifecycle management of a mobile retail and service business. In traditional mobile shops and service centers, critical operations such as inventory tracking, sales records, device servicing, warranty management, technician coordination, and customer service requests are often handled manually or through disconnected systems. This fragmented approach leads to several operational challenges including duplicate IMEI entries, lost warranty claims, delayed repair tracking, poor technician workload distribution, difficulty in fraud detection, and inefficient customer service management.

To address these challenges, a comprehensive web-based Mobile Shop Management System has been developed and implemented. The system integrates device inventory management, sales processing, repair tracking, warranty administration, technician job assignment, service request handling, and ownership transfer management into a unified platform. It provides real-time visibility of device status throughout its lifecycle, enables accurate IMEI-based tracking to prevent fraud, automates warranty expiry monitoring, streamlines repair workflows from customer request to technician completion, and maintains complete audit trails for compliance and analytics.

The system implements role-based access control with five distinct user types, each with specific permissions and responsibilities. Administrators manage user approvals, system monitoring, fraud detection, and generate comprehensive reports across all operations. Shop Owners handle inventory management, process sales transactions, assign customers to purchased devices, track stock levels, and generate sales invoices. Service Center managers coordinate repair operations, assign jobs to technicians, manage warranty claims, track parts inventory, and monitor service quality metrics. Technicians receive job assignments, update repair status in real-time, request required parts, and document completed work. Customers can track their device warranty status, submit service requests, view repair history, download invoices, and initiate ownership transfers for resale scenarios.

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Chapter 1
Introduction

The mobile phone industry started in the early 1970s, but it became widely popular and accessible to the general public during the 1990s. Over the years, the industry has seen rapid growth and continuous innovation, leading to the development of smartphones with advanced features. Today, mobile phones are not just communication devices but powerful tools used for business, education, online transactions, entertainment, and social connectivity. They have become an essential part of daily life, and it is difficult to imagine modern society without them.

Mobile shop businesses play a very important role in this growing industry. They act as a direct link between mobile manufacturers and customers. These shops are responsible for selling new devices, providing repair and maintenance services, and offering aftersales support. Customers depend on mobile shops not only for purchasing devices but also for solving technical issues and getting guidance about products. Despite the technological growth in mobile devices, many mobile shops still follow traditional methods of managing their business, such as maintaining handwritten records or using basic spreadsheets. These methods are time consuming, less accurate, and difficult to manage as the business grows.

The operations of a mobile shop are complex and involve multiple interconnected activities. A shop owner must handle different tasks at the same time to ensure smooth functioning of the business. These activities include:

1. Procurement of mobile devices and accessories from suppliers
2. Managing and updating inventory records
3. Performing sales transactions and billing
4. Recording and maintaining customer information
5. Managing product warranties and service records
6. Handling repair services and tracking device status

Each of these activities plays a crucial role in maintaining business efficiency and customer satisfaction. If any of these processes are not managed properly, it can lead to confusion, delays, and financial loss.

Apart from selling mobile phones, mobile shops also deal in various accessories and related products to increase their income and attract more customers. These include:

1. Protective covers and screen guards
2. Chargers, cables, and power banks

3. Earphones, headphones, and memory cards
4. Smart accessories like smartwatches and wireless earbuds

These additional products help shop owners generate extra revenue and improve customer satisfaction by providing all required items in one place.

Mobile shop businesses generate income from multiple sources. The main sources of income include the sale of new mobile devices, repair and maintenance services, and accessories. In addition, shop owners may also earn through SIM card activations, extended warranty services, and software related services such as unlocking or updating devices. However, along with income, there are also several expenses involved in running a mobile shop. These expenses include the cost of purchasing inventory, rent for the shop, salaries of employees, cost of spare parts for repairs, electricity and internet bills, marketing expenses, and government related fees. Proper management of both income and expenses is necessary to maintain profitability.

Managing all these operations manually creates several challenges for mobile shop owners. One of the major challenges is tracking each device using its unique IMEI number. Manual entry of IMEI numbers can lead to errors, duplication, or missing records. Customer information is also often recorded incorrectly or may be lost, which creates problems during warranty claims or aftersales service. Warranty tracking becomes difficult without proper records, and shop owners may not be able to verify whether a product is still under warranty.

Financial management is another area were mobile shop owners face difficulties. Calculating daily sales, monthly profits, expenses, and pending payments manually is time consuming and prone to errors. It becomes difficult to analyse business performance or make informed decisions. Lack of proper records also affects tax calculations and compliance with government regulations.

To run a mobile shop successfully, it is very important to maintain accurate and up to date records of all business activities. These include:

1. Inventory details and stock levels
2. Sales transactions and billing records
3. Customer information and purchase history
4. Repair status and service records
5. Warranty details and expiry dates
6. Financial data including income and expenses

Maintaining these records manually is not efficient, especially as the business grows in size and complexity.

Therefore, there is a strong need for an integrated Mobile Shop Management System that can handle all these operations in a centralized and automated manner. Such a system helps in reducing human

errors, saving time, and improving overall efficiency. It allows shop owners to easily track inventory, manage customer data, monitor repairs, and generate financial reports. It also helps in improving customer service by providing quick and accurate information.

Chapter 2

Literature Survey

A literature survey is an important part of system development because it helps us study the methods and technologies already used in the industry and understand the real problems faced by users. For our Mobile Shop Management System, the literature survey focuses on how existing mobile retail and service platforms manage device inventory, IMEI tracking, sales transactions, customer registration, warranty processing, repair coordination, technician assignment, and financial record keeping.

Through this study, we learned that mobile shop management platforms must handle multiple interconnected operations efficiently:

Customer/User Side: Customers should be able to register their purchased devices, view warranty status, submit service requests, track repair progress, download invoices, and initiate ownership transfers without confusion.

Shop Owner Side: Shop owners should be able to add device inventory with IMEI validation, process sales transactions, register customers, activate warranties, track stock levels, generate invoices, and maintain digital sales records.

Service Center Side: Service centers should be able to receive repair requests, assign jobs to technicians, track repair status, manage spare parts inventory, process warranty claims, and monitor service quality metrics.

Technician Side: Technicians should be able to view assigned jobs, update repair progress, request required parts, document completed work, and communicate repair status efficiently.

Admin Side: Administrators should be able to approve new shops/service centers, monitor all transactions, detect fraudulent activities through IMEI tracking, generate comprehensive reports, and manage user access across the platform.

We reviewed and compared different mobile shop management systems (standalone POS software, inventory management tools, repair tracking apps, warranty management platforms, and integrated retail solutions). This helped us understand the standard workflow of mobile retail businesses and identify critical limitations such as duplicate IMEI entries, warranty fraud, poor repair tracking, manual record errors, lack of technician coordination, difficulty in fraud detection, and absence of lifecycle visibility.

This survey inspired us to design a system that is comprehensive, secure, and efficient with complete

IMEIbased tracking throughout the device lifecycle and significantly reduced manual effort.

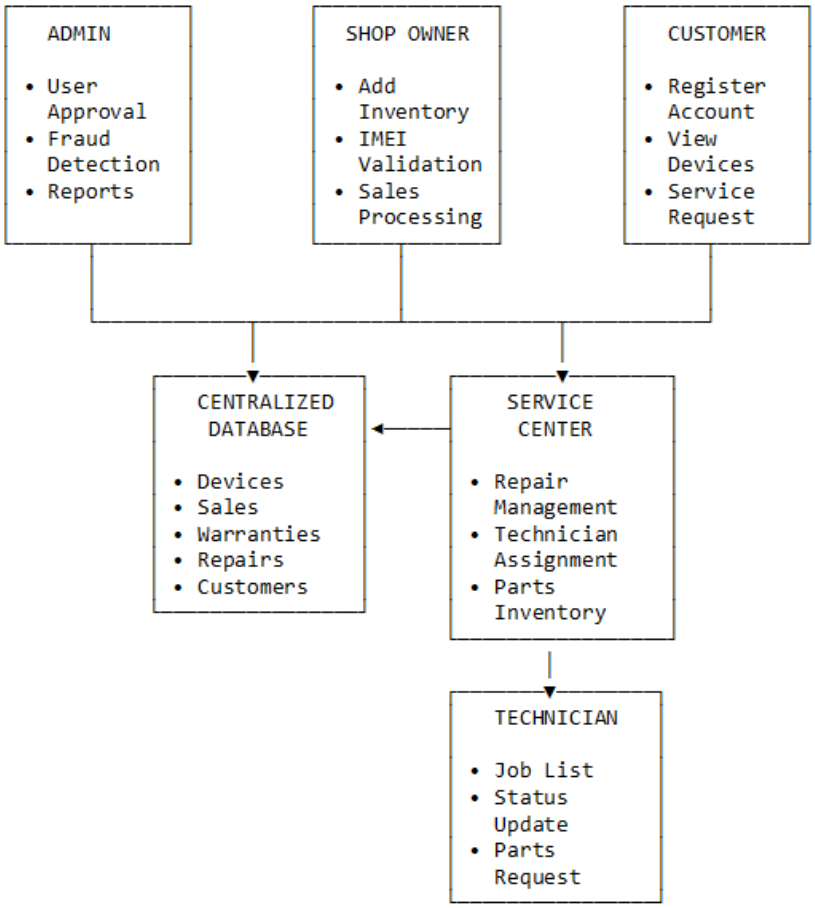

Figure 2.1: Complete Mobile Shop Management System Flow

Mobile shop management systems follow a comprehensive lifecycle approach to manage devices from procurement to eventual resale or disposal. Different modules handle different stakeholder responsibilities to ensure smooth operation and prevent issues like IMEI duplication, warranty fraud, and repair tracking failures.

Device Lifecycle Status Flow

Once a device enters the inventory, it moves through the following lifecycle stages:

IN_STOCK → SOLD → UNDER_WARRANTY → REPAIR_REQUESTED →
UNDER_REPAIR → REPAIRED → WARRANTY_EXPIRED → TRANSFERRED

Figure 2.2: Device Lifecycle Status Flow in Mobile Shop System

Repair Request Status Lifecycle

When a customer submits a repair request, it progresses through multiple status stages:

OPEN → ASSIGNED → IN_PROGRESS → PARTS_PENDING →
COMPLETED → CLOSED → DELIVERED

2.1 Detailed Process of Mobile Shop Management System

2.1.1 User Registration and Role Based Authentication

In mobile shop management platforms, authentication is the first step for accessing role specific features. The system supports multiple user types with different access levels.

1. Users create accounts using basic details (Name, Email, Phone, Password, Role).
2. Shop owners and service centers must provide additional business credentials (Shop Name, License Number, Address).
3. The system verifies user credentials at login and assigns appropriate dashboard access.
4. Admin approval is required for shop owner and service center registrations to prevent unauthorized access.
5. Customer registrations can be instant or linked to device purchase transactions.
6. After login, each user role accesses a personalized dashboard with rolespecific functionalities.

Importance: Rolebased authentication ensures secure access control, prevents unauthorized operations, and maintains proper audit trails for compliance and fraud detection.

2.1.2 Device Inventory Management with IMEI Validation

Shop owners must maintain accurate inventory records with unique IMEI tracking for each device.

Key features required: Device entry with mandatory fields: IMEI, Brand, Model, Color, Storage, Purchase Price, Selling Price IMEI validation (15digit numeric format check) Duplicate IMEI detection across the entire database to prevent fraud IMEI normalization (removing spaces, slashes, special characters) Stock status tracking (Available, Sold, Reserved, Defective) Bulk import capability for large inventory uploads Barcode/QR code generation for quick device lookup

Importance: Accurate IMEI tracking is critical for warranty validation, theft prevention, ownership verification, and fraud detection. Each device must be uniquely identifiable throughout its lifecycle.

2.1.3 Sales Transaction Processing

When a customer purchases a device, the system must record comprehensive transaction details.

Sales process workflow: 1. Shop owner searches available inventory by IMEI or model 2. Customer details are captured (name, email, phone, address) 3. Sale price, payment method, and invoice number are recorded 4. System automatically links device to customer profile 5. Device status changes from “In Stock” to “Sold” 6. Warranty is automatically activated with start date and

expiry calculation 7. Digital invoice is generated with PDF export capability 8. Transaction is recorded for financial reporting

Common Problem: If customer information is recorded incorrectly or incompletely, it becomes difficult to process warranty claims, handle service requests, or verify ownership during resale.

Importance: Structured sales processing ensures proper customer device linkage, enables warranty activation, generates financial records, and provides audit trails for compliance.

2.1.4 Warranty Activation and Tracking (Critical Step)

Warranty management is one of the most critical functions in mobile shop systems. This prevents warranty fraud and ensures legitimate claims are processed efficiently.

Warranty workflow: Warranty is automatically activated at the time of sale with start date recorded Expiry date is calculated based on warranty period (typically 624 months) System tracks warranty status: Active, Expiring Soon (within 30 days), Expired During repair requests, system automatically verifies warranty validity Automated alerts notify customers about upcoming warranty expiry Extended warranty purchases can be recorded separately

Common Problem: If warranty dates are not tracked accurately, it leads to disputes during repair claims, revenue loss from fraudulent claims, and customer dissatisfaction when legitimate claims are rejected.

Importance: Accurate warranty tracking prevents fraud, improves customer trust, reduces disputes, and ensures proper claim processing.

2.1.5 Service Request Submission and Tracking

Customers must be able to submit repair requests and track their status in realtime.

Service request workflow: 1. Customer logs in and navigates to “My Devices” 2. Selects the device requiring service 3. Provides problem description and urgency level 4. Submits request to admin/service center 5. System validates device ownership and warranty status 6. Request is assigned a unique tracking number 7. Status updates are visible to customer throughout repair process

Importance: Structured service request handling improves customer experience, enables proper tracking, reduces communication gaps, and ensures accountability.

2.1.6 Repair Assignment and Technician Coordination

Service centers must efficiently assign repair jobs to available technicians based on workload and expertise.

Assignment workflow: Service center receives repair request from admin or directly from customer Manager reviews device issue and warranty status Available technicians are displayed

with current workload Job is assigned to appropriate technician Technician receives notification and job details System tracks assignment timestamp and expected completion time

Common Problems: Uneven workload distribution leads to delays and technician burnout Lack of skillbased assignment results in longer repair times Poor tracking causes confusion about job status

Importance: Efficient technician assignment improves repair turnaround time, balances workload, ensures quality repairs, and enhances customer satisfaction.

2.1.7 Repair Status Updates and Parts Management

Technicians must update repair progress and request required parts during service.

Repair tracking features: Status updates: Diagnosis Started → Parts Required → Under Repair → Testing → Completed Parts request module with part name, quantity, and estimated cost Actual repair cost tracking vs. estimated cost Photo upload capability for documenting issues and completed work Time tracking for labor cost calculation Customer communication for approval of additional charges

Importance: Realtime status updates provide transparency, enable better customer communication, help manage parts inventory, and improve service quality.

2.1.8 Invoice Generation and Financial Reporting

Digital invoice generation and financial tracking are essential for business operations.

Invoice requirements: Autogenerated invoice numbers with sequential tracking Customer and device details Itemized pricing: Device cost, accessories, taxes, discounts Payment method and transaction reference PDF export and email/WhatsApp sharing capability Reprint functionality for lost invoices

Financial reporting needs:

Daily/monthly/yearly sales reports Revenue by brand, model, or price range Repair income vs. sales income breakdown Outstanding warranty claims Profit margin analysis

Importance: Digital invoices reduce paperwork, improve transparency, enable accurate financial tracking, support tax compliance, and provide business intelligence for decisionmaking.

2.1.9 Ownership Transfer and Device Resale

When customers resell their devices, ownership must be transferred properly to the new owner.

Transfer workflow: 1. Current owner initiates transfer request 2. Provides new owner's contact information 3. System verifies device ownership and warranty status 4. Transfer request is reviewed

by admin 5. New owner receives notification to accept transfer 6. Upon acceptance, device is linked to new owner profile 7. Remaining warranty is transferred (if applicable) 8. Transfer record is maintained for audit trail

Importance: Proper ownership transfer prevents fraud, maintains device history, enables warranty continuity, and supports secondhand market operations.

2.1.10 Fraud Detection and IMEI Verification

Admin monitoring and fraud prevention mechanisms are critical for system integrity.

Fraud detection features: Duplicate IMEI submissions are blocked automatically Crosschecking IMEI against stolen device databases (if integrated) Identifying suspicious patterns (multiple devices sold to same customer quickly) Monitoring warranty claims frequency per customer Tracking unusually high return rates for specific shops Flagging devices with multiple ownership transfers in short time

Importance: Fraud detection protects business reputation, prevents financial losses, ensures legal compliance, and maintains customer trust.

2.2 Study of Existing Systems

Existing mobile shop management systems range from basic spreadsheet tracking to sophisticated cloudbased platforms. The current market includes:

1. **PointofSale (POS) Systems** – Focus primarily on sales transactions and billing
2. **Inventory Management Software** – Track stock levels but lack repair and warranty modules
3. **Repair Shop Software** – Manage repairs but don't integrate with sales and inventory
4. **Warranty Management Platforms** – Track warranties separately without device lifecycle context
5. **AllinOne Retail Solutions** – Attempt comprehensive management but often expensive and complex

Key Observations from Existing Mobile Shop Platforms

- Most systems provide inventory tracking, but IMEI validation is often weak or absent
- Sales modules exist but rarely integrate warranty activation automatically
- Repair tracking modules exist separately, requiring manual data reentry
- Technician assignment is mostly manual without workload balancing

Table 1.Comparison of Existing Mobile Shop Management SystemsTable

System Type	**IMEI Tracking**	**Sales Processing**	**Warranty Management**	**Repair Tracking**	**Technician Assignment**
Basic POS (Marg, Busy)	Manual entry, no validation	Invoice generation only	Manual records	Not available	Not applicable
Inventory Software (Zoho Inventory)	Stock tracking only	Basic billing	Not integrated	Not available	Not applicable
Repair Shop Apps (RepairDesk)	Manual IMEI entry	Invoice for repairs only	Manual warranty check	Status tracking available	Manual assignment
Mobile Retail Solutions (Lightspeed)	Basic tracking	POS integration	Separate warranty module	Not integrated	Not available
Proposed System	Automated IMEI validation	Integrated warranty activation	Automated tracking with expiry alerts	Complete lifecycle tracking	Automated assignment

Table 2: Limitations of Existing Mobile Shop Management Systems

Sr. No.	Area	Limitations / Problems
1	IMEI Tracking	Duplicate IMEI entries allowed; no validation format check; manual entry errors
2	Warranty Management	Manual tracking; missed expiry dates; warranty fraud difficult to detect
3	Repair Coordination	No centralized tracking; status updates require phone calls; poor customer communication
4	Technician Management	Manual a assignment; uneven workload distribution; no performance tracking
5	Customer Experience	No selfservice portal; lack of realtime status updates; difficult to access device history
6	Financial Reporting	Basic sales reports; no repair income tracking.
7	Ownership Transfer	No systematic transfer.
8	Fraud Detection	No IMEI crossreferencing; stolen device sales possible; warranty

		claim fraud undetected

2.4 Problem Identification

After studying the existing systems, the following problems were clearly identified for all stakeholder groups:

Table 3: Problem Identification for All Stakeholders

Stakeholder	**Problem Identified**
Shop Owner Problems	Difficulty in tracking device inventory accurately due to manual IMEI entry and lack of validation
	Risk of duplicate IMEI entries leading to stock discrepancies and fraud
	Manual warranty activation process causes delays and errors in warranty period calculation
	Lack of integration between sales and warranty modules requires duplicate data entry
	Difficulty in generating comprehensive financial reports covering sales, repairs, and warranties
	No systematic way to track device lifecycle from procurement to resale
Service Center Problems	Poor coordination between shop owners and service centers for repair requests

	Manual technician assignment leads to workload imbalance and delays
	Difficulty in tracking repair status and communicating updates to customers
	Parts inventory management is manual and errorprone
	Warranty verification during repairs requires manual record checking
	No performance metrics for individual technicians
Technician Problems	Receiving job assignments through phone calls or manual notes
	Lack of visibility into job queue and expected workload
	Difficulty in requesting required parts systematically
	No digital record of completed work and customer approvals
Customer Problems	No visibility into warranty status and expiry dates
	Difficulty in submitting service requests and tracking repair progress

	Lost invoices cannot be retrieved easily
	Ownership transfer during device resale is complicated and undocumented
	Lack of trust due to no transparency in repair status and pricing
Admin Problems	Difficulty in detecting fraudulent activities such as duplicate IMEI sales
	Manual approval process for new shops and service centers
	Lack of centralized visibility into all transactions across shops and service centers
	Inability to generate systemwide reports for business analytics
	No automated alerts for suspicious patterns or warranty fraud

2.5 Needs of System

Based on the problems identified above, the following table outlines the functional requirements and needs of the proposed Mobile Shop Management System:

Table 4: Needs of the Proposed Mobile Shop Management System

Sr. No.	User / Role	Need of System
1	Shop Owner	IMEI validation with duplicate detection to prevent fraud and stock errors
2	Shop Owner	Integrated sales processing with automatic warranty activation
3	Shop Owner	Realtime inventory tracking
4	Shop Owner	Digital invoice generation with PDF export and email/SMS delivery
5	Shop Owner	Comprehensive financial reports covering all revenue streams and expenses
6	Service Center	Centralized repair request management with automated workflow
7	Service Center	Intelligent technician assignment based on workload and expertise
8	Service Center	Realtime repair status tracking with

		customer notification capability
9	Admin	Centralized dashboard with visibility.
10	Service Center	Integrated parts inventory management with lowstock alerts
11	Service Center	Automated warranty verification during repair intake
12	Technician	Digital job queue with prioritybased task list
13	Technician	Simple status update interface accessible via mobile devices
14	Technician	Digital parts request system with approval tracking
15	Customer	Selfservice portal to view purchased devices and warranty status
16	Customer	Easy service request submission with realtime tracking
17	Customer	Digital invoice access and download anytime

2.6 Problem Definition

Managing a mobile shop business through an application is challenging because the system must track devices individually by IMEI, process sales transactions, activate warranties, coordinate

repairs across multiple service centers, assign technician workloads, verify ownership, detect fraud, and provide realtime visibility to all stakeholders.

Chapter 3

Scope of the Project

This chapter defines the scope of the Mobile Shop Management System and explains what functions are included in the project. The system is designed to replace manual handling of mobile shop operations such as device inventory tracking, IMEI validation, sales processing, warranty management, repair coordination, technician assignment, and ownership transfer management.

It also defines userwise responsibilities so that Admin, Shop Owner, Service Center, Technician, and Customer can work in a structured and coordinated way.

The scope focuses on: Improving daytoday mobile shop and service center operations Eliminating IMEI duplication and fraud through automated validation Providing realtime device lifecycle visibility from purchase to resale Automating warranty activation, tracking, and expiry notifications Streamlining repair workflows with technician assignment and status tracking Enabling customer selfservice for device information and service requests Generating comprehensive reports for business decisionmaking Maintaining complete audit trails for compliance and dispute resolution

Table 5: System Functionalities According to Users

Sr. No.	Functionality	Admin	Shop Owner	Service Center	Technician	Customer
1	Login and authentication	Yes	Yes	Yes	Yes	Yes
2	User approval and management	Yes	No	No	No	No
3	View all devices across system	Yes	Own inventory	Related repairs	Assigned jobs only	Own devices only

			only	only		
Sr. No.	**Functionality**	**Admin**	**Shop Owner**	**Service Center**	**Technician**	**Customer**
4	Add device inventory with IMEI	No	Yes	No	No	No
5	IMEI validation and duplicate detection	Automatic	Automatic	Automatic	No	No
6	Process sales transactions	No	Yes	No	No	No
7	Generate and download invoices	Yes	Yes	Yes	No	Yes (own)
8	Activate and track warranties	Automatic on sale	Automatic on sale	View status	No	View own
9	Submit service requests	No	No	No	No	Yes
10	Receive and assign repair requests	Yes	Yes (forward to SC)	Yes	No	No
11	Assign jobs to	No	No	Yes	No	No

	technicians					
Sr. No.	**Functionality**	**Admin**	**Shop Owner**	**Service Center**	**Technician**	**Customer**
12	View and update repair status	Yes (all)	View related	Yes (all assigned)	Yes (own jobs)	Yes (own)
13	Request and manage spare parts	No	No	Yes	Yes	No
14	Track technician workload	Yes	No	Yes	View own	No
15	Initiate ownership transfer	No	No	No	No	Yes
16	Approve ownership transfer	Yes	No	No	No	Accept transfer
17	View warranty expiry alerts	Yes	Yes	Yes	No	Yes (own)
18	Fraud detection and alerts	Yes	No	No	No	No
19	Generate systemwide	Yes	Own shop only	Own center only	View own performance	No

	reports					

3.1 Functional Scope

The functional scope of the project covers the complete mobile device lifecycle from inventory procurement to eventual ownership transfer or disposal.

3.1.1 User Management and RoleBased Access Control

The system implements multirole user management with approval workflows.

1. **User Registration:** Support registration for all five user types with rolespecific fields
2. **Admin Approval:** Shop owners and service centers require admin approval before activation
3. **Role Assignment:** Assign appropriate access permissions based on user role
4. **Profile Management:** Users can update their profile information and credentials
5. **Status Management:** Admin can activate, block, or deactivate user accounts
6. **Audit Trail:** Track all user actions for security and compliance

3.1.2 Device Inventory Management with IMEI Validation

The system maintains accurate device inventory with unique IMEIbased tracking.

1. **Device Entry:** Add devices with complete details (IMEI, brand, model, color, storage, price)
2. **IMEI Validation:** Automatically validate 15digit IMEI format and normalize entry
3. **Duplicate Detection:** Prevent duplicate IMEI entries across entire database
4. **Stock Status Tracking:** Monitor availability status (In Stock, Sold, Reserved, Defective)
5. **Inventory Filters:** Search and filter by brand, model, price range, or status
6. **Low Stock Alerts:** Notify shop owners when inventory falls below threshold

3.1.3 Sales Transaction Processing

Sales management handles the complete transaction workflow from device selection to invoice generation.

1. **Customer Selection:** Search existing customers or create new customer profiles
2. **Device Selection:** Choose available devices from current inventory

3. **Transaction Recording:** Capture sale price, payment method, transaction date
4. **Automatic Warranty Activation:** Calculate and activate warranty period based on device type
5. **CustomerDevice Linking:** Associate purchased device with customer account
6. **Invoice Generation:** Generate professional PDF invoices with all transaction details
7. **Status Update:** Automatically change device status from “In Stock” to “Sold”
8. **Financial Recording:** Record transaction in sales reports and analytics

3.1.4 Warranty Management and Tracking

Warranty management is a critical component for customer satisfaction and fraud prevention.

1. **Automatic Activation:** Warranty starts automatically upon sale with proper date recording
2. **Expiry Calculation:** Calculate warranty end date based on device category (624 months)
3. **Status Tracking:** Monitor warranty status (Active, Expiring Soon, Expired)
4. **Expiry Alerts:** Send automated notifications 30 days before warranty expiration
5. **Claim Verification:** Automatically verify warranty validity during repair requests
6. **Claim History:** Maintain complete warranty claim history per device

3.1.5 Service Request Management

Customers can submit repair requests and track their progress in realtime.

1. **Request Submission:** Customers describe device issues and select urgency level
2. **Device Validation:** Verify device ownership and warranty status automatically
3. **Request Routing:** Forward requests to appropriate service center
4. **Tracking Number:** Generate unique reference number for each request
5. **Status Visibility:** Customers view realtime status updates
6. **Communication:** Send notifications on status changes via email/SMS
7. **Request History:** Maintain complete service request history per device.

3.1.6 Repair Workflow and Technician Assignment

Service centers coordinate repair operations from intake to completion.

1. **Repair Intake:** Receive requests from customers or shop owners
2. **Initial Assessment:** Review device issue, warranty status, and estimated cost
3. **Technician Selection:** View available technicians with current workload

4. **Job Assignment:** Assign repair to appropriate technician based on skills and availability

3.1.7 Parts Inventory and Procurement

Service centers maintain spare parts inventory for efficient repairs.

1. **Parts Catalog:** Maintain database of common spare parts with pricing
2. **Stock Tracking:** Monitor parts availability and consumption
3. **Parts Requests:** Technicians request parts for specific repair jobs
4. **Request Approval:** Service center approves and allocates parts

3.1.8 Ownership Transfer Management

The system handles device ownership changes during resale scenarios.

1. **Transfer Initiation:** Current owner starts transfer to new buyer
2. **New Owner Details:** Capture new owner's contact information
3. **Verification:** Validate current ownership and device status
4. **Transfer Approval:** Admin reviews and approves transfer request
5. **Acceptance Workflow:** New owner accepts transfer via email/SMS link

3.1.9 Fraud Detection and Security

Admin monitors system activities to detect and prevent fraudulent operations.

1. **Duplicate IMEI Detection:** Block attempts to register same IMEI multiple times
2. **Pattern Analysis:** Identify suspicious transaction patterns
3. **HighFrequency Alerts:** Flag customers with unusually frequent warranty claims
4. **Audit Logs:** Maintain detailed logs of all system activities

3.2 In Scope and Out of Scope

Table 6: Scope Boundary

Sr. No.	Area	In Scope	Out of Scope
1	Device Inventory	IMEI based tracking, duplicate detection, stock management	IoT based realtime location tracking of physical devices
2	Sales Processing	Transaction recording, invoice generation, warranty activation	Integration with external payment gateways for online payments
3	Warranty Management	Activation, tracking, expiry alerts, claim verification	Direct integration with manufacturer warranty databases
4	Repair Workflow	Request handling, technician assignment, status tracking	AIbased automatic fault diagnosis or repair cost estimation
5	Technician Management	Job assignment, workload balancing, performance tracking	Biometric attendance or advanced HR management
6	Customer Portal	Device viewing, service requests.	Mobile app development

Sr. No.	**Area**	**In Scope**	**Out of Scope**
7	Financial Reports	Sales and repair revenue, basic profit analysis	Full accounting ERP integration with ledgers and tax filing
8	Fraud Detection	IMEI duplicate checking, pattern analysis	Integration with national stolen device databases or law enforcement systems
9	Ownership Transfer	Transfer workflow, approval, warranty migration	Blockchainbased ownership verification or smart contracts

3.3 User Requirements

3.3.1 Admin

The Admin is responsible for overall system supervision, fraud monitoring, and regulatory compliance.

Core Requirements:

1. **User Management:** Approve/reject shop owner and service center registrations
2. **System Monitoring:** View all transactions across shops, service centers, and customers
3. **Fraud Detection:** Monitor IMEI duplication attempts and suspicious patterns
4. **Transfer Approvals:** Review and approve device ownership transfer requests
5. **Comprehensive Reporting:** Access systemwide analytics and business intelligence
6. **Alert Management:** Receive notifications for critical events and suspicious activities

7. **Data Quality:** Ensure data integrity and consistency across all module

3.3.2 Shop Owner

Shop owners manage device inventory, process sales, and coordinate with service centers.

Core Requirements:

1. **Inventory Management:** Add, update, and track device inventory with IMEI validation
2. **Sales Processing:** Create sales transactions with customer registration
3. **Invoice Generation:** Generate and print/email professional invoices
4. **Warranty Activation:** Automatically activate warranties at point of sale
5. **Stock Monitoring:** View realtime inventory levels and lowstock alerts
6. **Customer Database:** Maintain customer records and purchase history
7. **Service Coordination:** Forward customer service requests to service centers
8. **Sales Reports:** View daily, monthly, and yearly sales performance
9. **Financial Tracking:** Monitor revenue, profit margins, and pending payments

3.3.3 Service Center

Service center managers coordinate repair operations, assign technicians, and manage parts inventory.

Core Requirements:

1. **Repair Queue Management:** View incoming repair requests with priority indicators
2. **Technician Assignment:** Assign jobs based on technician availability and workload
3. **Warranty Verification:** Automatically verify warranty validity for incoming repairs
4. **Parts Management:** Handle parts inventory, requests, and procurement
5. **Cost Estimation:** Calculate estimated repair costs including parts and labor

6. **Customer Communication:** Send status updates to customers automatically

3.3.4 Technician

Technicians handle assigned repair jobs and update progress in realtime.

Core Requirements:

1. **Job List:** View all assigned jobs with device and issue details
2. **Priority Queue:** See urgent jobs highlighted for immediate attention
3. **Job Details:** Access complete customer information, device specifications.
4. **Status Updates:** Update repair status as work progress (Diagnosis, Under Repair).
5. **Time Tracking:** Record time spent on each repair task
6. **Cost Recording:** Enter actual parts and labor costs
7. **Performance View:** See personal performance metrics (jobs completed, average time)

3.3.5 Customer

Customers need selfservice access to device information, warranty status, and repair tracking.

Core Requirements:

1. **Account Access:** Secure login to view personal dashboard
2. **Device Portfolio:** View all purchased devices with warranty status
3. **Warranty Tracking:** See warranty start date, expiry date, and days remaining
4. **Service Requests:** Submit repair requests with issue description
5. **Repair Tracking:** Track realtime status of submitted service requests
6. **Invoice Access:** Download invoices for all purchases and repairs
7. **Transfer Initiation:** Initiate ownership transfer when selling device

3.4 Non Functional Scope

The system also addresses quality attributes essential for practical deployment and user satisfaction.

Usability: Intuitive user interface following modern design principles Responsive design working on desktop, tablet, and mobile browsers Consistent navigation across all modules Clear error messages and validation feedback Minimal training requirement for basic operations

Scalability: Database architecture supporting growth to 100,000+ devices Modular code structure allowing feature additions Efficient indexing for fast searches even with large data Horizontal scaling capability for future traffic growth

Maintainability: Modular code organization following MVC/component patterns Clear documentation for all major functions Standardized coding conventions Version control using Git Easy deployment and configuration management

Compatibility: Crossbrowser support (Chrome, Firefox, Safari, Edge) Mobile browser compatibility for onthego access PDF generation compatible with all major readers Data export in standard formats (CSV, Excel)

3.5 Expected Outcomes of Project Scope

By implementing the above scope, the project is expected to deliver significant improvements across operational, financial, and customer satisfaction dimensions:

Operational Improvements: Elimination of IMEI duplication through automated validation, preventing stock discrepancies and fraud Reduced manual effort by automating warranty activation, expiry tracking, and customer notifications Faster transaction processing with integrated saleswarrantycustomer workflows.

Financial Benefits: Accurate profit tracking across sales and repair revenue streams Reduced warranty fraud through systematic verification and claim tracking Better inventory management minimizing dead stock and stockouts Transparent pricing reducing customer disputes and refund requests Datadriven decisions based on comprehensive financial analytics

Customer Experience: Selfservice access to device information, warranty status, and repair tracking Transparency in repair status and cost estimation Faster service through optimized technician assignment Proactive communication via automated status updates and warranty expiry alerts Easy ownership transfer when reselling devices

Chapter 4

Methodology

This chapter explains the methodology used to plan, design, and develop the Mobile Shop Management System. The objective of the methodology is to ensure that development remains structured, measurable, and aligned with actual business needs.

The proposed system replaces manual mobile shop operations with a webased process for managing device inventory, IMEI validation, sales transactions, warranty activation, repair coordination, technician assignment, ownership transfers, and comprehensive reporting.

To keep development disciplined, the project follows the Waterfall Model of Software Development Life Cycle (SDLC), where each phase is completed and reviewed before moving to the next.

4.1 SDLC Approach Followed

4.1.1 Waterfall Model

The Waterfall Model is selected because the project requirements are clear and can be implemented phase by phase. Each phase produces verified outputs before the next phase begins, ensuring systematic progress and better control.

Phases followed:

1.Requirement Analysis:

Requirement analysis was performed through structured observation, user interviews, and study of existing mobile shop operations to identify the actual needs of the business.

2.System Design:

In this phase, requirements are converted into architecture, process flows.

3.Implementation:

The system is developed by converting the design into code using appropriate technologies and tools. All modules like inventory and repair management are integrated and made functional.

4.Testing:

Testing is done to check errors, bugs, and performance issues in the system. It ensures that all features work correctly and meet user requirements.

5.Deployment

Deployment is the process of making the system available for actual use in the real environment. It involves installing, configuring, and running the system for users.

6.Maintenance:

Maintenance involves updating the system, fixing bugs, and improving performance.

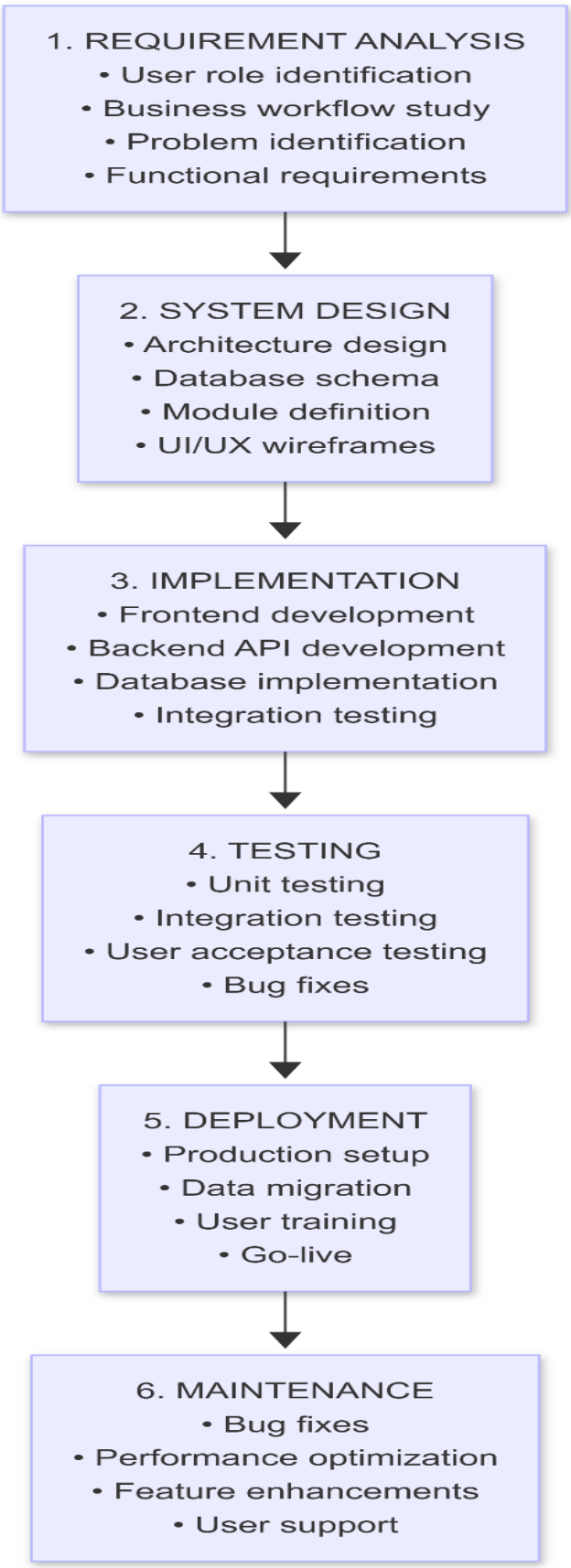

Figure 4.1: SDLC Waterfall Model Sequential Phase Flow

4.2 Requirement Analysis

Requirement analysis was performed through structured observation, user interviews, and study of existing mobile shop operations to identify the actual needs of the business.

Research Activities: Discussion with mobile shop owners and service center managers Observation of manual device tracking and sales workflows Study of common failures in warranty management and repair coordination Analysis of IMEI duplication and fraud issues Mapping required modules with user roles

4.2.1 Admin Requirements

Admin requires complete system oversight, fraud monitoring, and regulatory compliance.

Key Requirements: Approve or reject new shop owner and service center registrations. Monitor all transactions across the entire system. Detect and flag IMEI duplication attempts in realtime. Review and approve device ownership transfer requests. Access comprehensive reports and analytics across all modules. Configure system parameters and user permissions. Receive automated alerts for suspicious activities. Maintain audit trails for compliance and dispute resolution. Generate systemwide performance reports. Manage user accounts (activate, block, deactivate).

4.2.2 Shop Owner Requirements

Shop owners handle inventory procurement, sales transactions, and customer registration.

Key Requirements: Add device inventory with automatic IMEI validation. Prevent duplicate IMEI entries during data entry. Process sales transactions with customer linking. Automatically activate warranties upon sale. Generate professional PDF invoices. Track inventory status (available, sold, reserved). View sales performance reports (daily, monthly, yearly). Search devices by IMEI, brand, model, or status. Forward customer service requests to service centers Monitor warranty expiry for customer followups. Track profit margins by device category.

4.2.3 Service Center Requirements

Service centers coordinate repair operations, assign technicians, and manage parts inventory.

Key Requirements: Receive repair requests from customers and shop owners. Verify warranty status automatically for incoming repairs. View available technicians with current workload. Assign repair jobs intelligently based on expertise and availability. Track repair status through multiple stages. Manage spare parts inventory and procurement. Approve parts requests from technicians. Estimate repair costs before starting work. Send status updates to customers automatically. Generate service performance reports (turnaround time, completion rate). Monitor individual technician performance metrics.

4.2.4 Technician Requirements

Technicians need simple, mobilefriendly access to their job queue and status updates.

Key Requirements: View assigned repair jobs with priority indicators. Access complete device and customer information for each job. Update repair status as work progresses. Request required spare parts through the system. Upload photos documenting issues and repairs. Record actual time spent and costs incurred. Add completion notes and recommendations. View personal performance metrics. Receive notifications for new job assignments. Mark jobs as completed once finished.

4.2.5 Customer Requirements

Customers require transparency, selfservice access, and realtime tracking capabilities.

Key Requirements: Register and maintain personal account. View all purchased devices in one place. Check warranty status and expiry dates for each device. Submit service requests with issue descriptions. Track repair status in realtime without phone calls. Download invoices for purchases and repairs. Receive automated notifications for warranty expiry. Initiate ownership transfer when selling devices. View complete service history per device. Accept ownership transfers from previous owners. Provide feedback on repair services

4.3 System Design Methodology

In this phase, requirements are converted into architecture, process flows, and modulelevel design. The proposed design includes four main components:

1. **Input Layer** : User interactions and data entry points
2. **Processing Layer** : Business logic and system modules
3. **Database Layer** : Data storage and management
4. **Output Layer** : Generated outputs and user interfaces

4.4 Functional Methodology by Module

4.4.1 Device Inventory and IMEI Validation Module

This module ensures accurate device tracking and prevents IMEI duplication fraud.

Methodology:

1. Shop owner enters device details including IMEI
2. System validates IMEI format (exactly 15 digits, numeric only)

Data Validation Rules: IMEI must be exactly 15 digits. IMEI must be numeric only. IMEI must be unique across all devices Brand, model, and price fields are mandatory. Selling price cannot be less than purchase price (warning shown)

4.4.2 Sales Transaction and Warranty Activation Module

This module processes sales and automatically activates warranties.

Methodology:

1. Shop owner searches for available device by IMEI or model
2. System displays device details and current status
3. Shop owner enters or selects customer information
4. System captures sale price, payment method, invoice number

Automatic Warranty Calculation: Flagship phones: 24 months warranty Midrange phones: 12 months warranty. Budget phones: 6 months warranty Feature phones: 6 months warranty Refurbished devices: 3 months warranty.

4.4.3 Service Request and Repair Workflow Module

This module manages the complete repair lifecycle from request to completion.

Methodology:

1. Customer logs in and views owned devices
2. Customer selects device requiring service
3. System verifies device ownership automatically
4. Customer provides problem description and urgency

Status Progression: OPEN → ASSIGNED → IN_PROGRESS → PARTS_PENDING (if needed) → TESTING → COMPLETED → CLOSED → DELIVERED

4.4.4 Technician Assignment and Workload Balancing Module

This module ensures fair distribution of repair jobs among technicians.

Methodology: 1. Service center receives new repair request 2. System displays list of all active technicians 3. For each technician, system shows: Current assigned jobs count Jobs in progress vs. pending Average completion time Specialization/expertise areas 4. Manager can sort technicians by workload (ascending) 5. System can autosuggest technician with lowest workload 6. Manager assigns job to selected technician 7. Technician's workload count increases automatically 8. Upon job completion, workload count decreases 9. Performance metrics are updated in realtime

Workload Calculation: Pending jobs: weight = 1.0 Inprogress jobs: weight = 1.5 (higher priority) Partspending jobs: weight = 0.5 (waiting state) Total workload score = sum of weighted jobs.

4.4.5 Ownership Transfer and Verification Module

This module handles device resale scenarios with proper documentation.

Methodology:

1. Current owner logs in and selects device to transfer

2. System verifies current ownership in database 3. Owner provides new owner's email and contact information 4. System creates transfer request with status "PENDING_ADMIN" 5. Admin receives notification to review transfer 6. Admin verifies device history (no disputes, valid purchase) 7. Admin approves or rejects transfer with remarks 8. If approved, new owner receives email/SMS with acceptance link 9. New owner registers/logs in and accepts transfer 10. System updates device ownership to new owner 11. Remaining warranty is transferred automatically 12. Transfer record is permanently stored for audit trail 13. Both parties receive confirmation notifications

Transfer Validation Rules: Device must be fully paid (no pending dues) Device must not have open repair requests Device must not be reported stolen Original purchase must be at least 7 days old (cooling period) Admin approval required for all transfers.

4.4.6 Fraud Detection and Alert Module

This module proactively identifies suspicious patterns and prevents fraud.

Methodology:

1. System monitors all IMEI entries in real time

2. On duplicate IMEI attempt: Entry is immediately blocked Admin receives instant alert with shop owner details Audit log records attempted entry

3. System analyzes warranty claim patterns: Flags customers with >3 claims in 6 months Flags devices with multiple claims for same issue

4.4.7 Reporting and Analytics Module

This module provides business intelligence through comprehensive reports.

Methodology:

1. User selects report type from dashboard
2. User specifies date range and filters
3. System queries relevant database collections
4. System aggregates data based on report requirements

4.5 User Management Methodology

User management is implemented using role based access control (RBAC) with approval workflows.

Access Roles in System

1. **Admin** – System supervisor with complete access
2. **Shop Owner** – Manages inventory, sales, and customers
3. **Service Center** – Coordinates repairs and technicians
4. **Technician** – Handles assigned repair jobs
5. **Customer** – Self service portal access

Security Controls Applied

Authentication: Password hashing using bcryptjs (10 salt rounds) Secure session management with JWT tokens Password strength requirements (min 8 characters, mix of letters/numbers) Account lockout after 5 failed login attempts

Authorization: Role based access control enforced on every API endpoint Middleware check user role before allowing access Frontend components render based on user permissions Restricted access to sensitive modules (admin only features)

4.6 Process Methodology

The project defines a complete operational flow from device procurement to eventual ownership transfer. The flow handles both standard operations and edge cases such as warranty expiry.

4.7 Module Architecture Diagram

The system is divided into eleven interdependent modules. Each module feeds into the next, ensuring a traceable and controlled device lifecycle.

4.8 User wise Module Designs

4.8.1 Admin Module Design

Main Functionalities: User registration approval/rejection workflow Systemwide transaction monitoring dashboard IMEI duplication and fraud detection alerts Ownership transfer request approval Comprehensive analytics and reporting User account management (activate, block, deactivate) Audit log viewing and analysis System configuration and settings

Key Screens: 1. User Approval Queue 2. Fraud Detection Dashboard 3. Transfer Approval List 4. System Wide Analytics 5. Audit Trail Viewer.

4.8.2 Shop Owner Module Design

Main Functionalities: Device inventory management with IMEI validation Sales transaction processing Customer registration and management Invoice generation and printing Warranty activation (automatic) Sales performance reports Stock level monitoring Service request forwarding

to service centers

Key Screens: 1. Inventory Management (Add/Edit devices) 2. Sales Processing (Create sale transaction) 3. Customer Database 4. Invoice Generator 5. Sales Dashboard with charts.

4.8.3 Service Center Module Design

Main Functionalities: Repair request queue management Technician workload monitoring Job assignment to technicians Warranty verification for repairs Parts inventory management Parts request approval Repair status tracking Service performance analytics.

4.8.4 Technician Module Design

Main Functionalities: View assigned job list with priorities Access device and customer details Update repair status in real time Request required spare parts Upload progress photos Record actual time and cost Add completion notes View personal performance metrics.

4.8.5 Customer Module Design

Main Functionalities: View purchased device portfolio Check warranty status and expiry dates Submit service requests Track repair progress in real time Download invoices Initiate ownership transfer Accept incoming transfer requests View complete service history.

Chapter 5

Details of Designs, Working and Processes

5.1 System Diagram

5.1.1 System Flow Diagram

The Mobile Shop Management System manages the complete device lifecycle from inventory procurement to eventual ownership transfer or disposal. It eliminates manual errors by maintaining all operational data in one centralized system with real time IMEI validation and updates. Every action performed by any user is immediately reflected across all connected modules, ensuring consistency, traceability, and fraud prevention throughout the device lifecycle.

The system serves five user facing roles — Admin, Shop Owner, Service Center, Technician, and Customer — each interacting with the system in a defined and controlled manner. All data operations are handled through a MongoDB database that stores and retrieves information for every device entry, sales transaction, warranty activation, repair request, technician assignment, parts request, ownership transfer, and comprehensive report.

The system has five user facing roles and one central data management layer:

1. **Admin** — System oversight, user approvals, fraud detection, and comprehensive monitoring

2. **Shop Owner** — Device inventory management, sales processing, customer registration, and invoice generation

3. **Service Center** — Repair coordination, technician assignment, parts management, and warranty verification

4. **Technician** — Job queue management, repair status updates, parts requests, and completion documentation

5. **Customer** — Self service portal for device viewing, warranty tracking, service requests, and ownership transfers

6. **Database backed system services** — Realtime data storage, IMEI validation, duplicate detection, and retrieval

The diagram below shows how each actor interacts with the central Mobile Shop Management System and what outputs are produced:

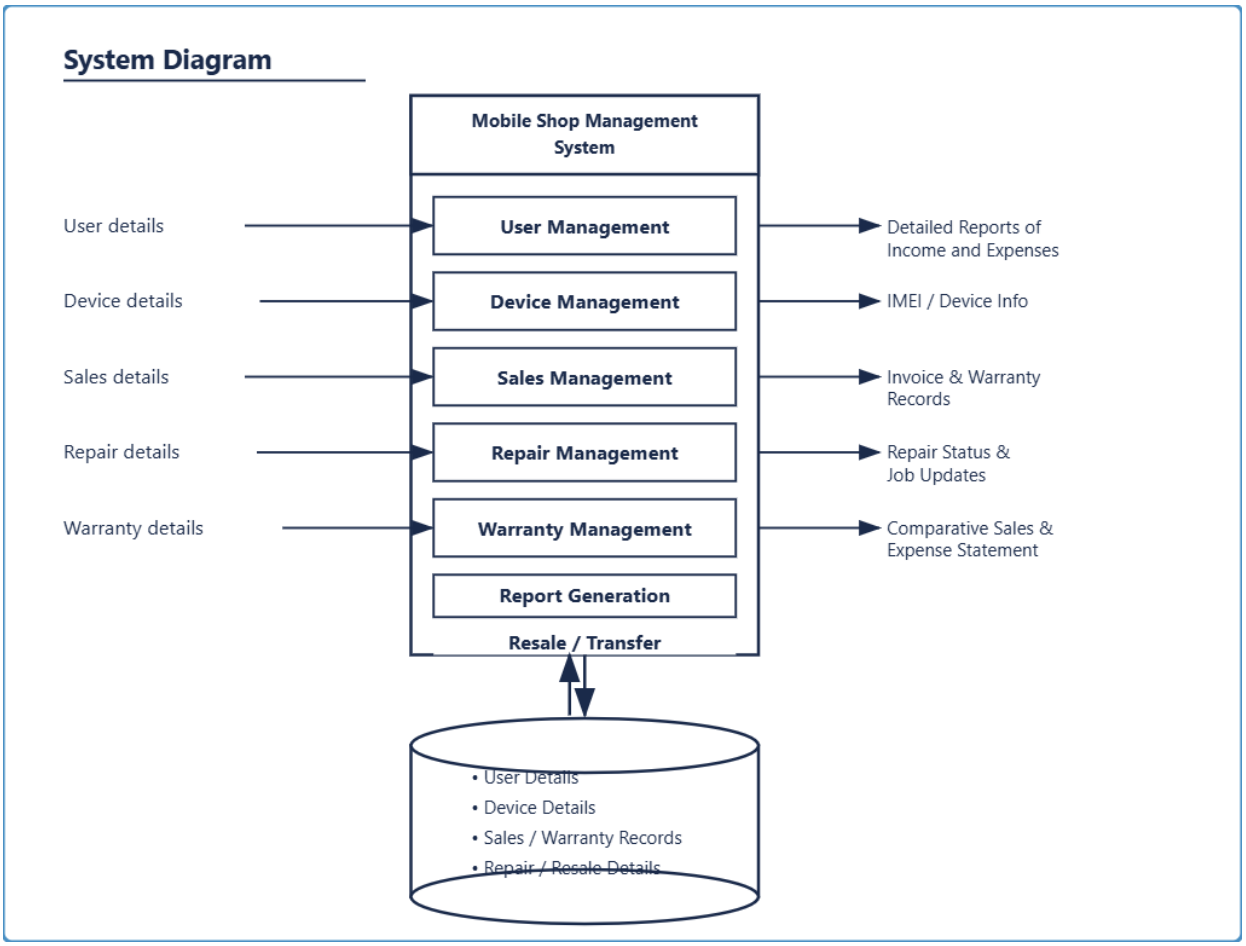

Figure 5.1: System Flow Diagram

5.1.2 Admin Module

The Admin Module provides centralized oversight, managing user approvals, fraud detection, and systemwide data integrity. It allows administrators to review IMEI entries, sales, warranties, and ownership transfers to ensure platform security and business compliance.

Core actions include approving registrations for shops and service centers, monitoring fraud alerts, and generating reports for business intelligence. This oversight is vital for maintaining a secure and trustworthy environment for all stakeholders.

Key features: Approve or reject user registrations Monitor all IMEI entries and detect duplicates Flag suspicious transaction patterns Approve device ownership transfers View comprehensive dashboards and analytics Manage user account status (active, blocked) Generate systemwide reports

5.1.3 Shop Owner Module

The Shop Owner Module facilitates inventory management, sales processing, and customer registration. Owners use this interface to register devices with IMEI validation, activate warranties automatically during sales, and generate professional PDF invoices, covering the entire retail lifecycle.

While restricted from system level settings, owners have full operational control over their specific store data. The module enforces strict validation rules to prevent duplicate IMEI entries and ensure accurate inventory tracking.

Key features: Add device inventory with IMEI validation Prevent duplicate IMEI entries automatically Process sales transactions with customer linking Automatically activate warranties upon sale Generate PDF invoices Track sales performance and profit margins Forward service requests to service centers

5.1.4 Service Center Module

The Service Center Module manages the repair lifecycle, including receiving requests, assigning technicians, and tracking parts inventory. Managers focus on workload balancing and warranty verification to ensure high quality service and timely delivery to customers.

Dedicated to service operations, this module optimizes the repair workflow. Managers can monitor technician performance and parts consumption through integrated service analytics without accessing retail sales data.

Key features: Receive repair requests from customers Verify warranty status automatically Assign jobs to available technicians Balance technician workload Manage spare parts inventory Approve parts requests from technicians Track repair status through all stages Generate service performance reports

5.1.5 Technician Module

The Technician Module offers a mobile friendly interface for managing assigned repairs. Technicians can view their queue, update job statuses in realtime, request spare parts, and document completed repairs with photos and notes.

This streamlined interface prioritizes efficient task execution. Technicians have direct access to their

workload and personal performance metrics, ensuring minimal administrative burden while keeping the system updated.

Key features: View assigned job queue with priorities Access device and customer details Update repair status as work progresses Request required spare parts Upload photos documenting repairs Record actual time and costs Add completion notes View personal performance metrics

5.1.6 Customer Module

The Customer Module provides a self service portal for users to manage their devices and track service progress. It enables transparent access to warranty details, repair history, and invoice downloads with minimal user effort.

For data privacy, customer access is strictly limited to their own purchase and service records. This module enhances the user experience by providing realtime updates and an easy way to initiate ownership transfers for resale.

Key features: View purchased device portfolio Check warranty status and expiry dates Submit service requests with issue description Track repair status in real time Download invoices (sales and repairs) Initiate ownership transfer for resale Accept incoming transfer requests View complete service history per device

5.2 Use Case Diagram

The Use Case Diagram identifies the system actors and maps them to the set of actions they are permitted to perform within the Mobile Shop Management System. It serves as a highlevel view of system functionality from the perspective of each user role.

The system has five major actors — Admin, Shop Owner, Service Center, Technician, and Customer. Each actor interacts with a distinct set of use cases based on their role. Admin has the widest access, covering user approvals and fraud monitoring. Shop Owner focuses on inventory and sales. Service Center manages repair coordination. Technician executes repair tasks. Customer has access to self service features.

This diagram is essential for understanding the boundary of each role and for ensuring that access control is correctly implemented in the system.

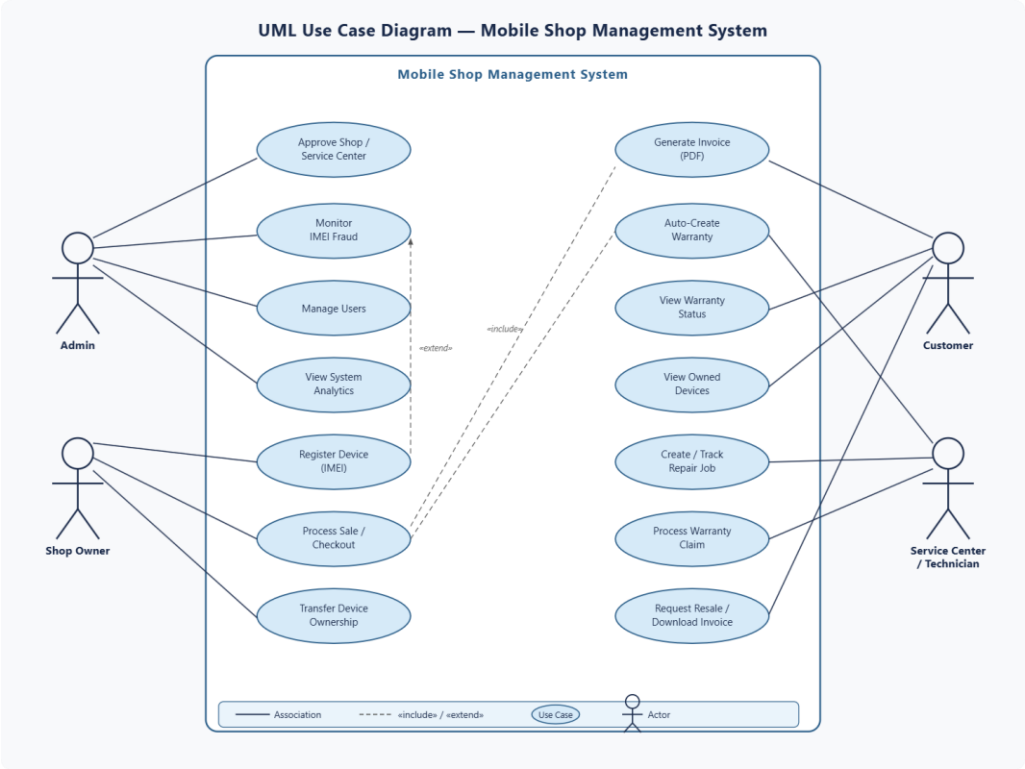

Figure 5.2: Use Case Diagram — System Actors and Rolewise Use Cases

5.2.1 Actor Responsibilities

Admin Activities

The Admin manages system security, user approvals, and fraud monitoring, generating reports that drive business decisions.

Responsibilities: Manage registrations, detect IMEI fraud, approve device transfers, and maintain systemwide audit trails.

Shop Owner Activities

Shop owners handle core retail tasks including inventory registration, sales recording, and professional invoicing.

Responsibilities: Maintain unique IMEI stock, process transactions, activate warranties, and monitor sales performance.

Service Center Activities

Service centers manage the repair lifecycle, ensuring efficient job distribution and parts inventory

control.

Responsibilities: Assign repairs to technicians, verify warranties, manage spare parts, and provide customer updates.

Technician Activities

Technicians perform physical repairs and provide real time digital status updates on their work queue.

Responsibilities: Execute repairs, request parts, document work with photos, and record labor time.

Customer Activities

Customers use the portal for transparent self service access to their device and service history.

Responsibilities: View warranty status, submit repair requests, track progress, and initiate ownership transfers.

5.3 Database Schema

The database is the backbone of the Mobile Shop Management System. It stores all operational data in a structured MongoDB format and supports the complete device lifecycle — from user authentication and device inventory with IMEI validation to sales transactions with warranty activation, repair tracking with technician assignment, parts management, ownership transfers, and comprehensive reporting.

The schema is designed around ten core collections, each representing a distinct operational concept. Relationships between collections are carefully defined through reference fields to ensure that data integrity is maintained across all operations. The database structure also supports advanced querying for fraud detection (duplicate IMEI checks), warranty expiry monitoring (automated alerts), technician workload analysis (job count per technician), and comprehensive analytics (sales trends, repair performance, profit margins).

Chapter 6

Results and Application

6.1 Result

This chapter presents the results of the developed Mobile Shop Management System through major web pages and their functional descriptions. The screenshots demonstrate how different users interact with the system for device lifecycle operations including inventory management, IMEI validation, sales processing, warranty tracking, repair coordination, and ownership transfers.

The implemented web application successfully executes: Role based authentication for five user types IMEI validation and duplicate detection Sales processing with automatic warranty activation Repair workflow with technician assignment Customer self service portal Fraud detection and admin monitoring Comprehensive reporting and analytics

6.2 Public Pages

Figure 6.1 shows the home page with navigation, hero section highlighting device lifecycle management, and feature cards explaining IMEI validation, warranty automation, and repair tracking.

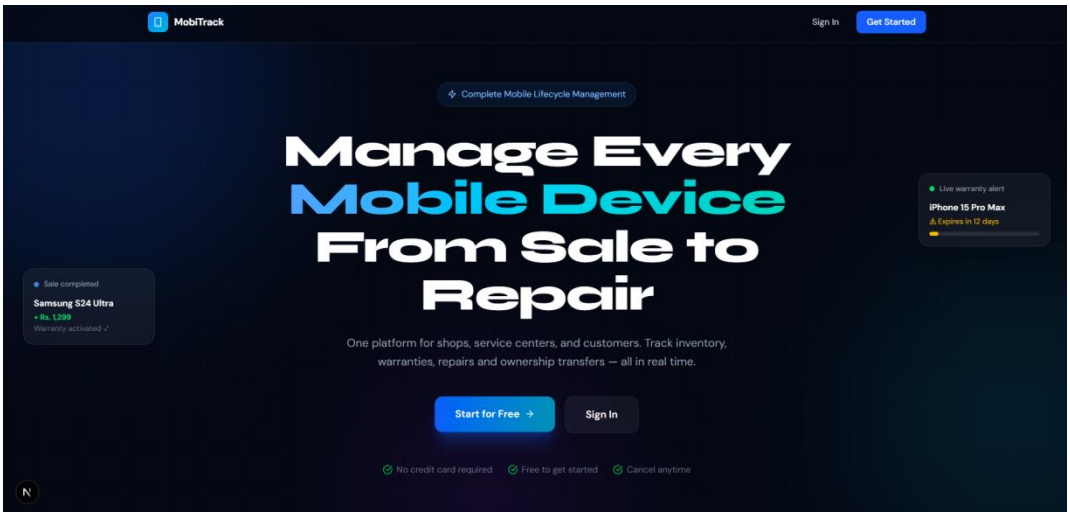

Fig. 6.1: Home Page Platform Introduction

Figure 6.2 shows the login page where users authenticate based on their role. Upon successful login,

the system redirects to rolespecific dashboards.

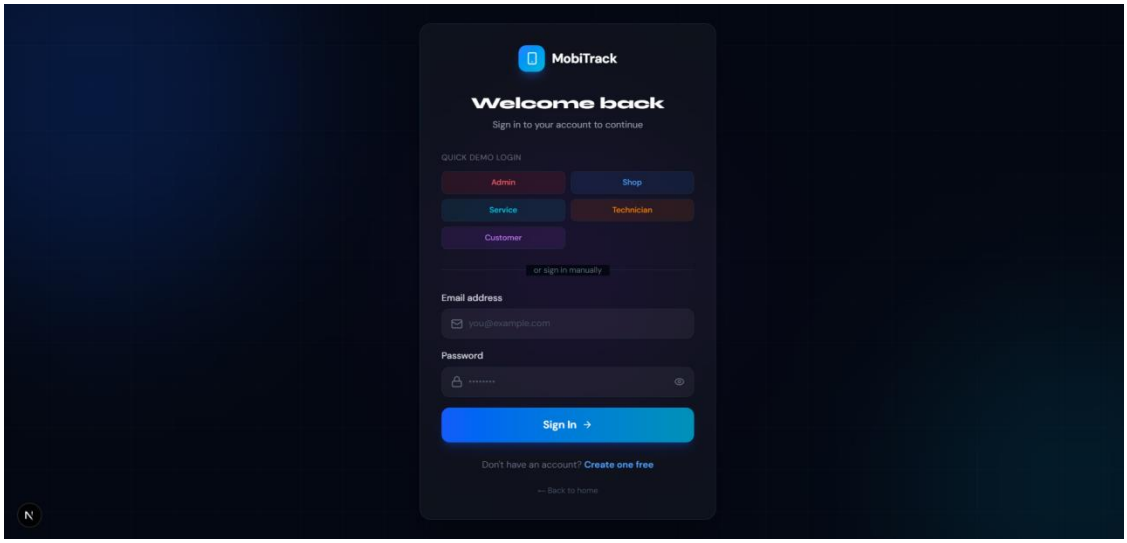

Fig. 6.2: Login Page Role Based Authentication

Figure 6.3 shows the registration page capturing user details, role selection, and rolespecific information. Shop owners and service centers require admin approval before activation.

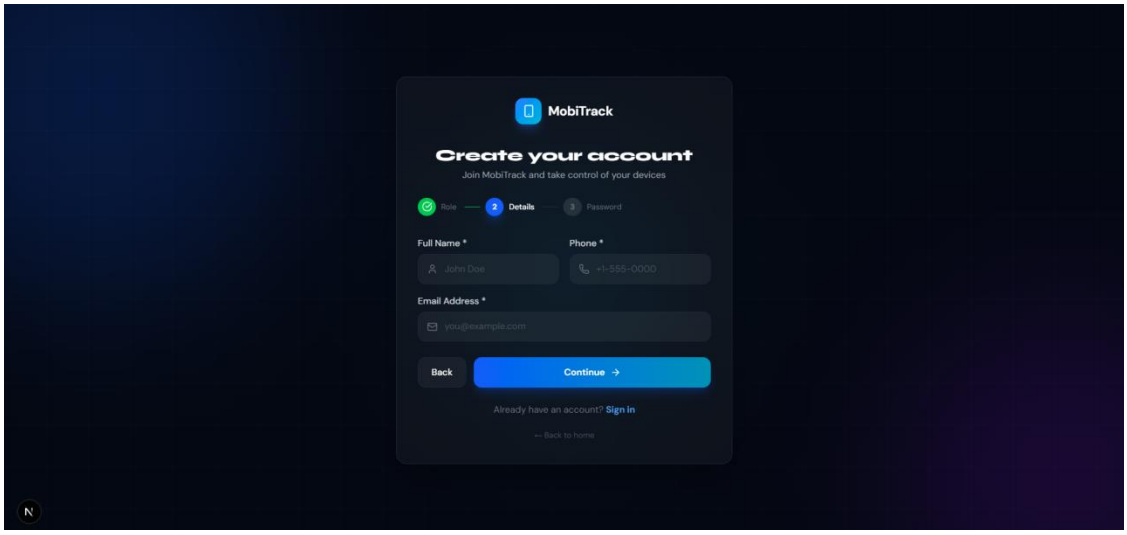

Fig. 6.3: Register Page User Onboarding with Role Selection

6.3 Customer Side Pages

Figure 6.4 shows the customer dashboard displaying all owned devices with warranty status, purchase dates, and quick action buttons.

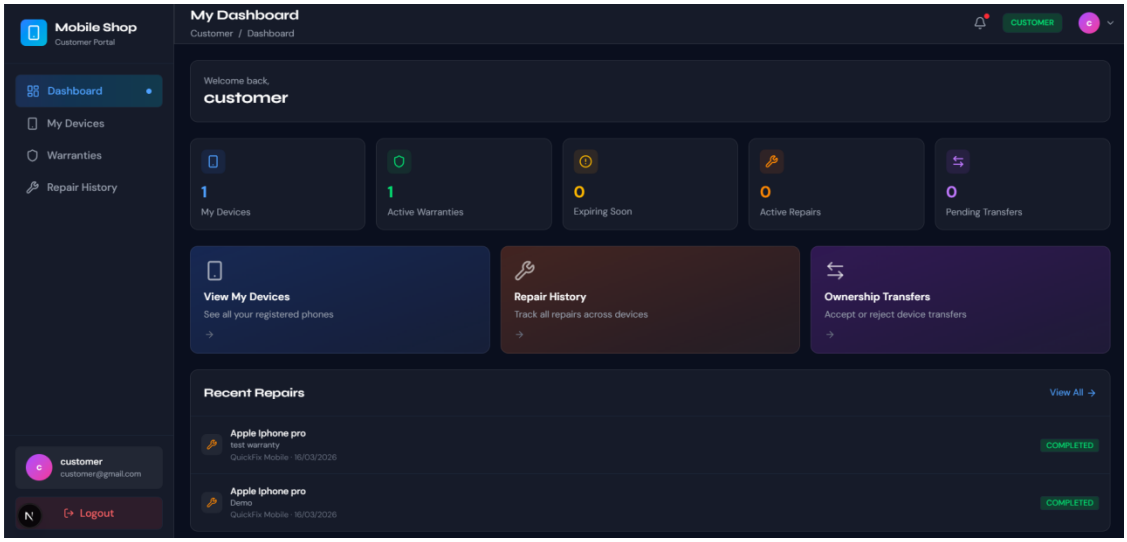

Fig. 6.4: Customer Dashboard Device Portfolio

Figure 6.5 shows complete device information including specifications, purchase details, warranty status, and service history.

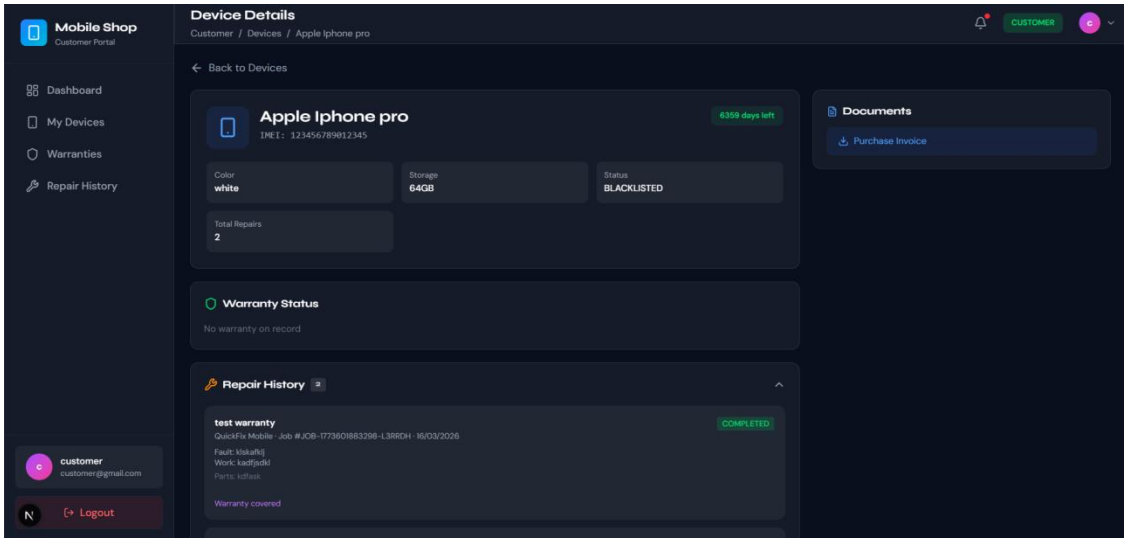

Fig. 6.5: Device Details Page Complete Device Information

Figure 6.6 shows the service request form with device selection, problem description, urgency level, and automatic warranty verification.

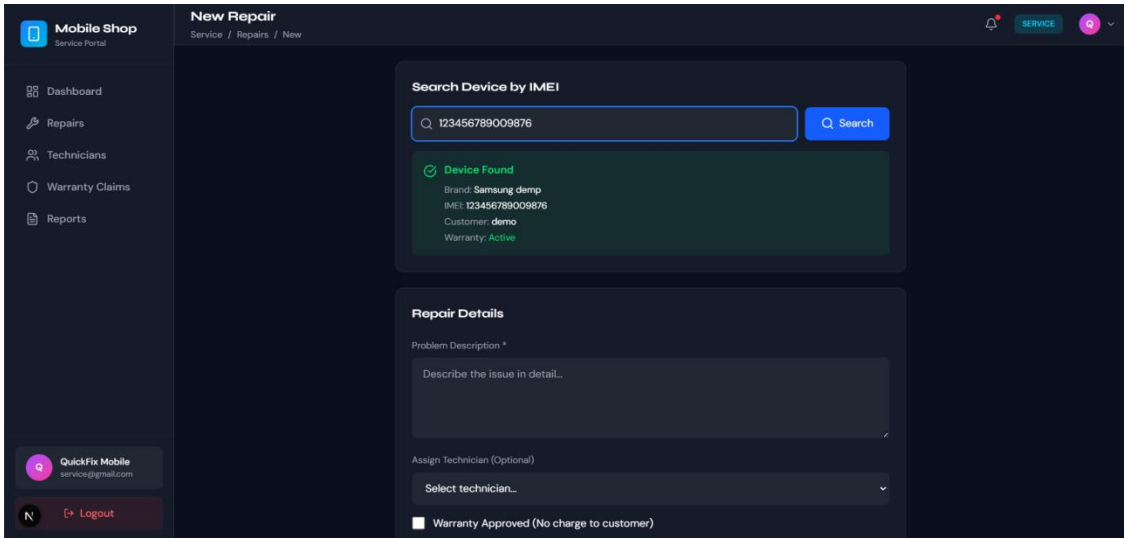

Fig. 6.6: Service Request Form Repair Request Submission

Figure 6.7 shows repair tracking with visual status progression, assigned technician, and realtime updates.

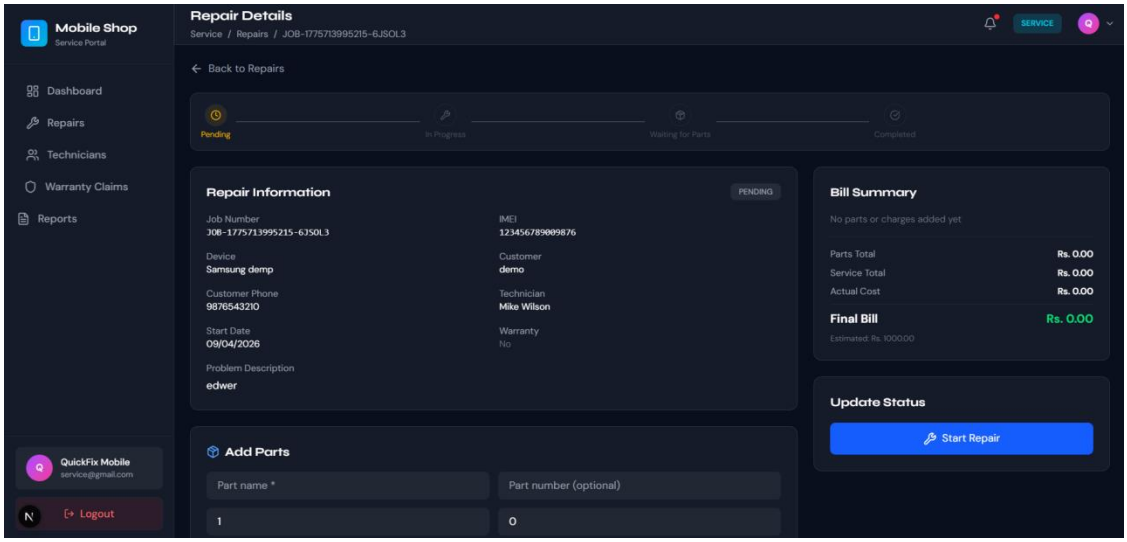

Fig. 6.7: Repair Tracking Page RealTime Status

Figure 6.8 shows the ownership transfer form for device resale with new owner details and transfer reason.

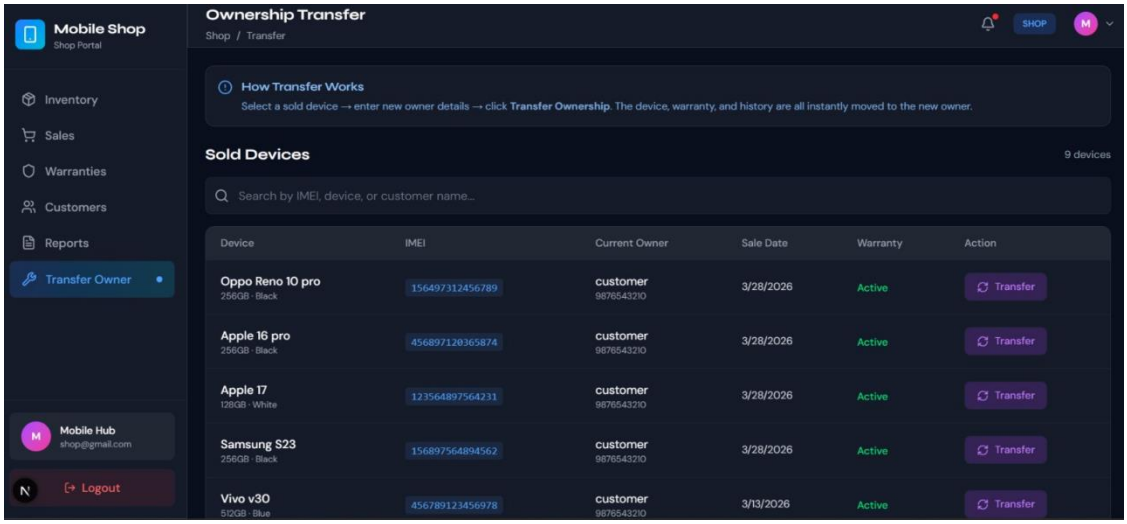

Fig. 6.8: Ownership Transfer Form Device Resale

6.4 Shop Owner Side Pages

Figure 6.9 shows business metrics including inventory count, sales revenue, top selling devices, and quick action shortcuts.

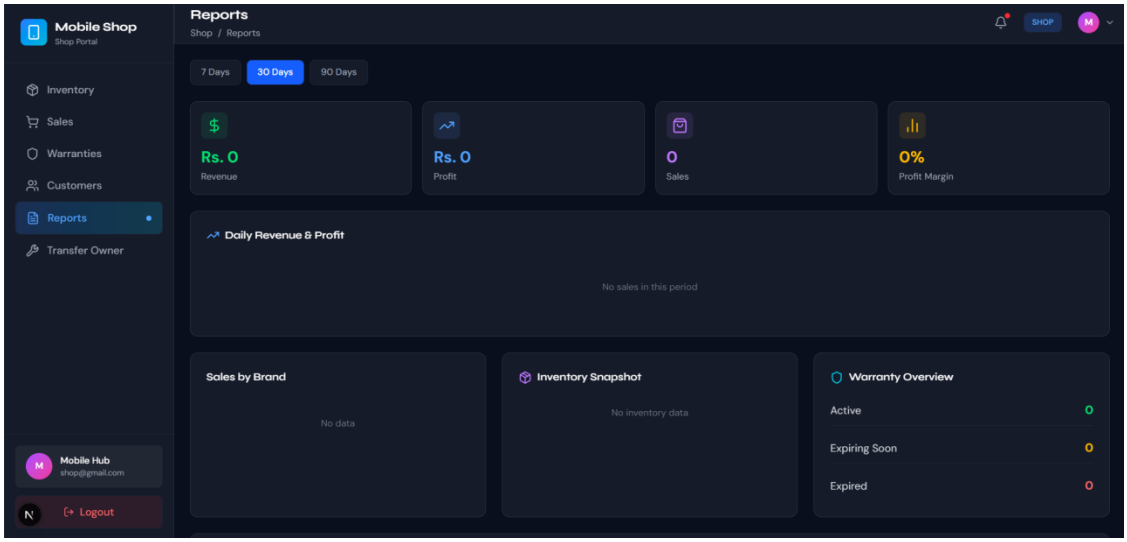

Fig. 6.9: Shop Owner Dashboard Business Metrics

Figure 6.10 shows device entry form with real time IMEI validation and duplicate detection.

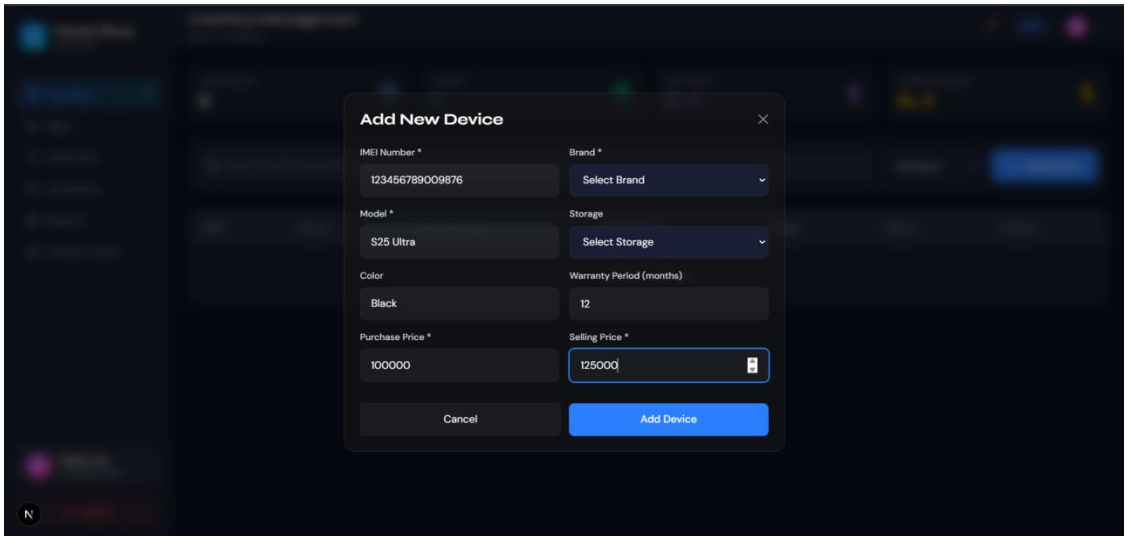

Fig. 6.10: Add Device Form IMEI Validation

Figure 6.11 shows sales form with device selection, customer details, and automatic warranty activation upon submission.

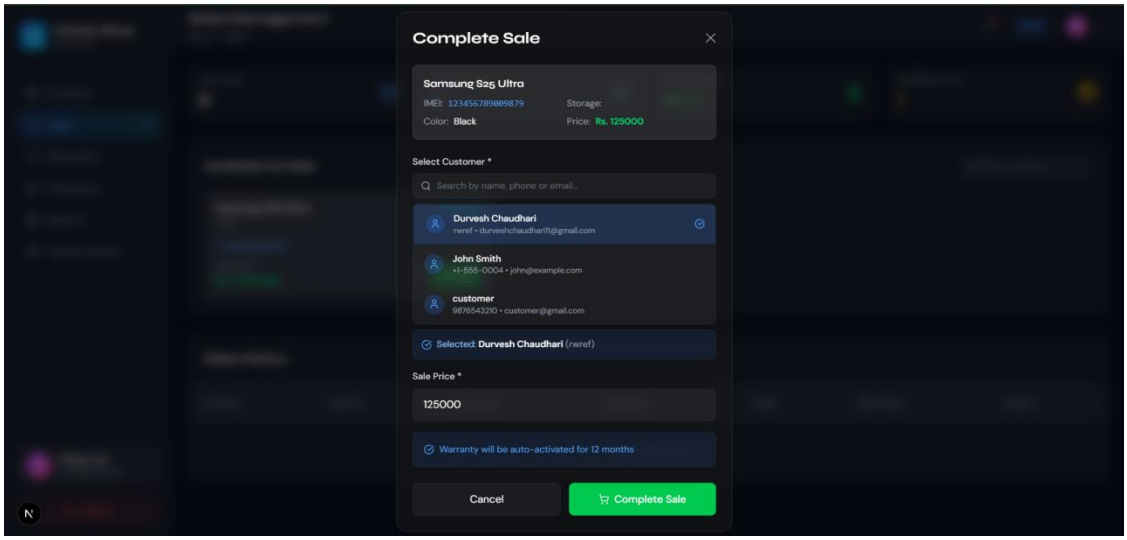

Fig. 6.11: Sales Transaction Auto Warranty Activation

Figure 6.12 shows the professionally formatted PDF invoice with all transaction details and warranty information.

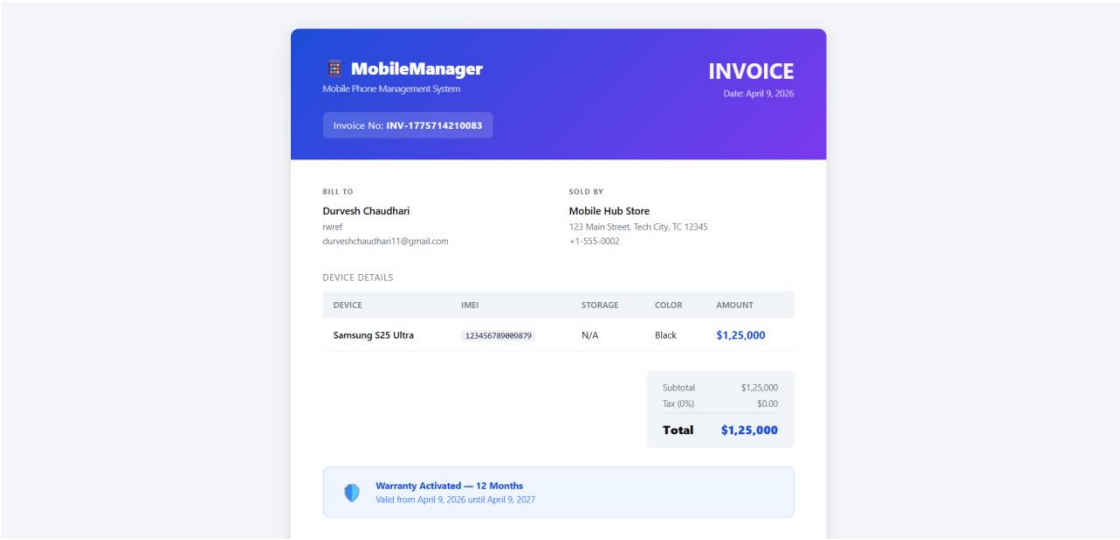

Fig. 6.12: PDF Invoice Professional Documentation

6.5 Admin Side Pages

Figure 6.13 shows systemwide metrics, fraud alerts, and pending approvals with visual analytics.

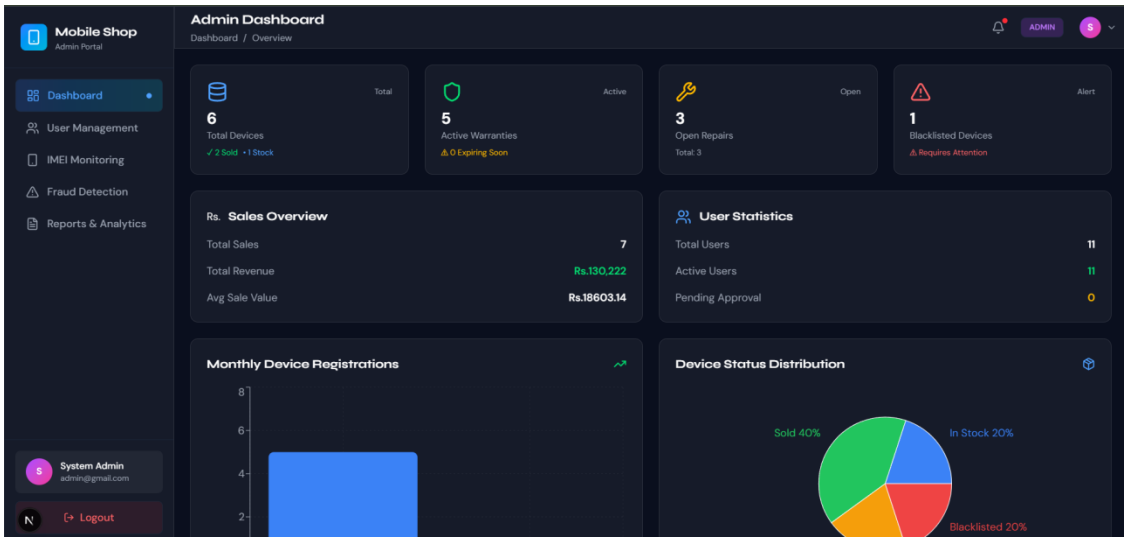

Fig. 6.13: Admin Dashboard System Monitoring

Figure 6.14 shows fraud alerts including duplicate IMEI attempts and suspicious patterns.

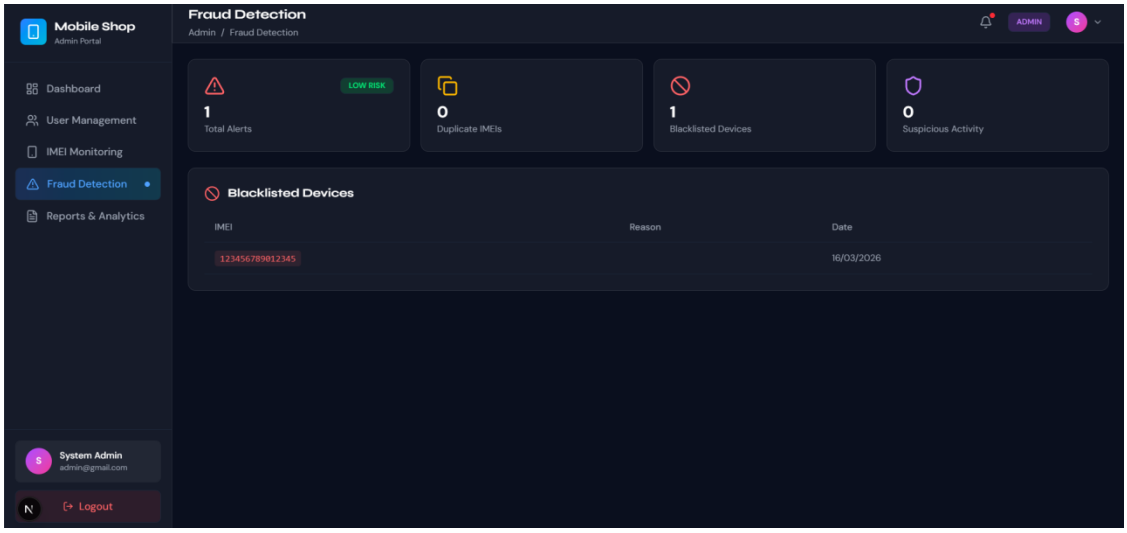

Fig. 6.14: Fraud Detection Security Monitoring

Figure 6.15 shows comprehensive reporting with filters and export options.

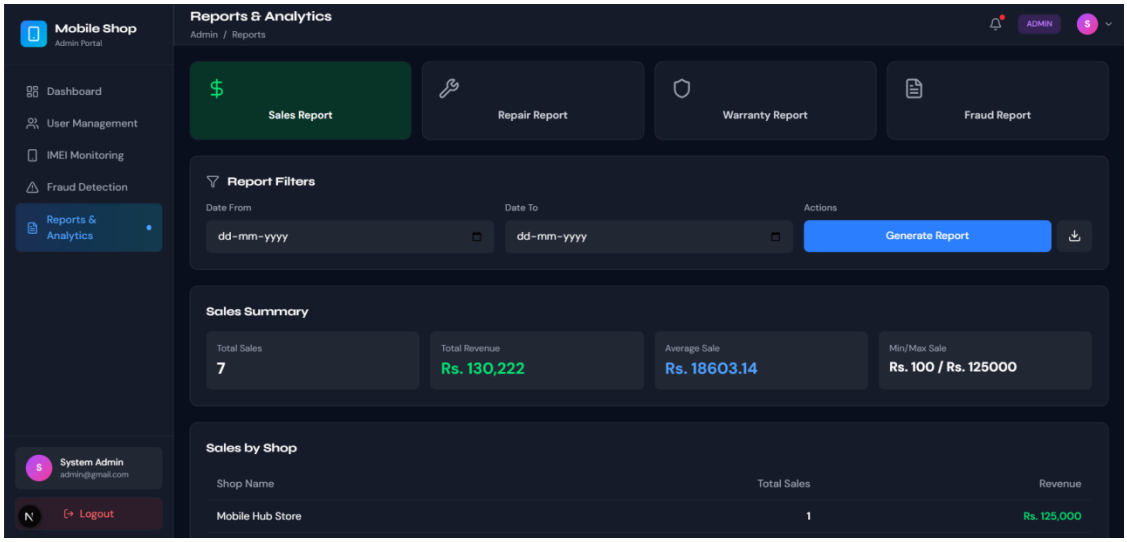

Fig. 6.15: System Reports Analytics

6.6 Application

The Mobile Shop Management System is applicable for: Small and medium mobile shops moving from manual IMEI tracking Multilocation retail chains requiring centralized management Service centers needing repair workflow coordination Franchise operations requiring fraud prevention oversight Businesses handling device trade ins and resale.

Practical applications: Device lifecycle tracking from procurement to resale IMEIbased fraud prevention and duplicate detection Warranty automation and expiry tracking Repair coordination with technician assignment Ownership transfer documentation financial intelligence.

6.7 Result Analysis

Improvements achieved: Eliminated IMEI fraud through real time duplicate detection Automated warranty management eliminating calculation errors Improved repair transparency with customer self service Balanced technician workload through intelligent assignment Systematic ownership transfers with audit trails Financial clarity separating sales and repair revenue Role based security with complete audit logs

Comparative Analysis:

Table 7: Comparative Analysis — Manual vs. Developed System

Parameter	**Manual Process**	**Developed System**
IMEI Tracking	Prone to duplicates	Autovalidation with duplicate detection
Warranty Management	Manual calculation	Automatic activation and alerts
Fraud Detection	Reactive	Proactive realtime monitoring
Repair Tracking	Phone calls	Realtime status portal
Technician Assignment	Manual, unbalanced	Workloadbased intelligent assignment
Ownership Transfers	Informal	Structured approval workflow
Financial Reporting	Manual compilation	Realtime dashboards
User Accountability	Low traceability	Complete audit logs

CHAPTER 7

Testing

7.1 Introduction

This chapter covers the testing activities performed for critical modules of the Mobile Shop Management System. Testing is essential because the system handles sensitive operations including IMEI validation, fraud prevention, warranty management, and financial transactions. If core modules like authentication, IMEI validation, or sales processing fail, the entire system becomes unreliable and vulnerable to fraud.

The objective of this testing phase is to verify that: Authentication works correctly for all five user roles (Admin, Shop Owner, Service Center, Technician, Customer) IMEI validation prevents duplicate entries and fraud Sales processing correctly activates warranties automatically Repair workflow assigns technicians and tracks status properly Ownership transfer requires proper approvals and verification.

Testing was performed using:

Manual Testing:

Functional validation of business logic and user workflows

Postman API Testing:

Backend endpoint validation with requestresponse verification

Browser DevTools:

Frontend validation and console error monitoring

The testing scope includes:

User authentication and rolebased access control IMEI validation and duplicate detection Sales transaction processing and warranty activation Service request submission and repair tracking Ownership transfer workflow and approvals.

7.2 Test Cases

This section documents all test cases designed and executed for critical modules of the Mobile Shop Management System. Test cases cover authentication, IMEI validation, sales processing, warranty management, repair workflow, and ownership transfers.

7.2.1 Comprehensive Test Cases

Table 8: Comprehensive Test Cases — Mobile Shop Management System (TCAUTH001 to TCREPORT003)

TC-ID	Test Scenario	Preconditions	Test Steps	Test Data	Expected Result	Status
TCAUTH001	Login with valid Admin credentials	Admin user exists in DB	Open login → Enter valid email and password → Click Login	Valid Admin credentials	Login successful; redirect to Admin dashboard	Pass
TCAUTH002	Login with valid Shop Owner credentials	Shop Owner exists in DB	Open login → Enter credentials → Click Login	Valid Shop Owner credentials	Login successful; redirect to Shop dashboard	Pass
TCAUTH003	Login with invalid password	User exists in DB	Enter valid email + wrong password	Wrong password	Error message shown; login blocked	Pass
TCAUTH004	Access protected route without login	User not logged in	Open protected URL directly in browser	Protected dashboard URL	Redirect to login page	Pass

TCIMEI001	Add device with valid unique IMEI	Shop owner logged in	Navigate to Add Device → Enter all fields with valid 15digit IMEI → Submit	Valid IMEI: 123456789012345	Device added successfully; status IN_STOCK	Pass
TCIMEI002	Add device with duplicate IMEI	Device with same IMEI exists	Enter device details with existing IMEI → Submit	Duplicate IMEI: 123456789012345	Error message: "IMEI already exists"; entry blocked	Pass
TCIMEI003	Add device with invalid IMEI format	Shop owner logged in	Enter IMEI with less than 15 digits → Submit	Invalid IMEI: 12345	Validation error: "IMEI must be 15 digits"	Pass
TCIMEI004	Add device with nonnumeric IMEI	Shop owner logged in	Enter IMEI with letters → Submit	Invalid IMEI: ABC123456789012	Validation error: "IMEI must be numeric"	Pass
TCIMEI005	IMEI normalization with spaces	Shop owner logged in	Enter IMEI with spaces → Submit	IMEI with spaces: "123 456 789 012 345"	System removes spaces; IMEI saved as 123456789012345	Pass

TCSALES001	Process sales transaction with valid data	Device in stock, customer exists	Select device → Enter customer info → Enter price → Submit	Valid device + customer	Sale recorded; device status changed to SOLD	Pass
TCSALES002	Automatic warranty activation on sale	Device sold successfully	Complete sales transaction → Check warranty record	Device sale completed	Warranty created with start date = sale date, end date = sale date + period	Pass
TCSALES003	Generate PDF invoice after sale	Sale completed	Complete sale → Click Generate Invoice	Valid sale ID	PDF invoice generated with all details	Pass
TCREPAIR003	Assign repair job to technician	Service request received	Service center views technicians → Select technician → Assign	Available technician	Job assigned; technician receives notification	Pass
TCREPAIR004	Technician updates repair status	Job assigned to technician	Technician logs in → Select job → Update status to IN_PROGRESS	Assigned job	Status updated; customer notified	Pass
TCREPAIR	Technician requests	Repair in progress	Technician clicks Request Parts → Enter part details	Part name and	Parts request created; sent to service	Pass

005	parts		→ Submit	quantity	center	
TCREPAIR006	Service center approves	Parts request pending	Service center reviews request → Click Approve	Valid parts request	Parts allocated; technician	Pass
TCREPORT001	Generate sales report for date range	Sales data exists	Navigate to Reports → Select date range → Generate Sales Report	Date: Last 30 days	Report shows sales count, revenue, profit	Pass
TCREPORT002	Generate repair performance report	Repair data exists	Navigate to Reports → Select Service Reports → Generate	Service center data	Report shows turnaround time, completion rate	Pass
TCREPORT003	Export report to PDF	Report generated	View report → Click Export as PDF	Any report	PDF file downloaded with all data	Pass

7.3 Test Summary

Testing of the Mobile Shop Management System confirms that all critical modules are stable and functioning as expected. The test cycle covered 35 test cases across authentication, IMEI validation, sales processing, warranty management, repair workflow, fraud detection, and reporting modules.

Chapter 8

Conclusion and Future Scope

8.1 Conclusion

The Mobile Shop Management System was developed to digitize and simplify the complete lifecycle management of mobile retail and service operations. The system replaces manual recordkeeping with a structured and centralized workflow for device inventory, IMEI validation, sales processing, warranty management, repair coordination, ownership transfers, and fraud detection.

The system significantly improves operational accuracy and service quality through:

1. IMEI based fraud prevention with automatic validation and real time duplicate detection.
2. Automated warranty management with expiry tracking and instant verification.
3. Intelligent repair workflow with systematic technician assignment and status tracking.
4. Complete lifecycle visibility from procurement to resale with audit trails.
5. Customer self service portal for transparent device and repair tracking.
6. Financial intelligence with comprehensive sales and repair analytics.
7. Role based security and complete audit logging for compliance.

By integrating all mobile shop workflows into one unified platform, the project reduces manual effort, prevents fraud, improves transparency, and supports scalable business growth.

8.2 Future Scope

The current system establishes a strong foundation for mobile shop lifecycle management. In future versions, the following enhancements can further improve performance and user experience.

Instant MultiChannel Notifications

Send automatic alerts through SMS, WhatsApp, and email for booking confirmation, warranty expiry reminders, repair status updates, payment receipts, and fraud detection alerts.

Mobile Application Development

Provide Android and iOS mobile apps or Progressive Web App (PWA) for staff, technicians, and customers to manage operations and track repairs from anywhere.

Online Payment Gateway Integration

Support integrated UPI, card, and wallet payments with automated receipt generation, refund processing, and payment reconciliation for both sales and repairs.

QR Code and Barcode Scanning

Use barcode or QR scanning to speed up device identification, reduce manual IMEI entry errors.

Chapter 9

References and Bibliography

9.1 References

To understand mobile shop operations, device lifecycle management, and practical software solutions, we reviewed project resources, industry standards, and technical documentation.

1. **Internal Project Repository and Source Code** Study of implemented modules for user authentication, IMEI validation, device inventory, sales processing, warranty management, repair workflow, technician assignment, ownership transfers, fraud detection, and analytics dashboards.
2. **API Testing Collection and Backend Documentation** Review of backend API endpoint design and requestresponse validation through project API documentation and test collections for sales, repairs, warranties, and transfers.
3. **Mobile Retail and Service Management Studies** Functional review of market systems and industry practices for device inventory tracking, IMEI validation, warranty lifecycle management, repair coordination, and service center operations.
4. **Next.js and MERN Stack Documentation** Use of official documentation for Next.js, React, Node.js, MongoDB, and associated libraries for implementation best practices and performance optimization.
5. **IMEI Standards and Device Identification** Review of international standards for IMEI format validation, duplicate detection methods, and mobile device identification systems.
6. **RoleBased Access Control and Security** Review of authentication methods, password hashing with bcryptjs, rolebased authorization, and protected routes in web applications.
7. **PDF Generation and Invoice Standards** Study of jsPDF library documentation and invoice formatting standards for generating professional business documents.
8. **Database Design and NoSQL Practices** MongoDB schema design patterns for document relationships, indexing strategies for performance, and query optimization techniques.

9.2 Bibliography

1. **Next.js Documentation. Available at: https://nextjs.org/docs**
2. **React Documentation. Available at: https://react.dev/**
3. **MongoDB Documentation. Available at: https://www.mongodb.com/docs/**
4. **Mongoose Documentation. Available at: https://mongoosejs.com/docs/**